MIL-P-46476C (PA)
20 May 1975
SUPERSEDING
MIL-P-46476B (MU)
15 September 1963

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M81
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of primer designated as Primer, Percussion, M81.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-P-223	- Powder, Black
MIL-P-20449	- Primer, Percussion, M61, Loading, Assembling and Packing
MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes

FSC: 1390

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

8848625 - Primer, Percussion, M81, Loading Assembly
8848888 - Box, Packing, Ammunition, for Primer, Percussion, M81
8848889 - Carton, Packing, Ammunition, for Primer, Percussion, M81

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Primer.-The primer shall comply with all requirements specified on Drawing (dwg.) 8848625 and with all requirements specified in applicable specifications.

3.3 Head assembly

3.3.1 Non-functioning.-The head assembly shall not function, when tested as specified in 4.3.1.1.

3.3.2 Functioning.-The head assembly shall function on the first attempt, when tested as specified in 4.3.1.2.

3.4 Primer assembly functioning.-The Primer assembly shall function, there shall be no audible hangfire and no part shall show evidence of rupture, break or become detached, when tested as specified in 4.3.2.

3.5 Moisture content of black powder.-The moisture content of the black powder at the loading station, at the time of loading, shall not exceed 0.5 percent. Testing shall be as specified in 4.3.3.

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3.6 Workmanship.-All parts and assemblies shall be clean and free of burrs, chips, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few small scratches. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.2 Inspection provisions

4.2.1 Inspection Lot Formation.-The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All materials submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirement. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

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a. Component parts bearing the same lot interfix number from one manufacturer.

b. Black powder from not more than one lot.

c. Percussion elements from not more than one lot.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Sampling plans shall be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Head, prior to assembling battery cup and firing plug (see dwg. 8848625)

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
101.	Percussion element missing, cocked or inverted	Visual
Minor:		
201.	Seal between percussion element assembly and head assembly missing or incomplete	Visual
202.	Evidence of poor workmanship (see 3.6)	Visual

4.2.2.2 Head, prior to assembling to body (see dwg. 8848625)

Categories	Defects	Method of Inspection
Critical:		
1.	Firing plug or battery cup above flush	Straight-edge/Visual

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Major:
101. Firing plug missing, cocked or inverted.....Visual
102. Flash hole in head completely obstructed .....Visual
103. Battery cup loose .....Manual/Visual

4.2.2.3 Body, prior to assembly of liner (see dwg. 8848623 covering a detail of dwg. 8848625)

Categories	Defects	Method of Inspection
Critical:		
	1. Any flash hole missing from body	Gage
	2. Body cracked or split	Visual
Major: None defined		
Minor:		
	201. Evidence of poor workmanship (see 3.6)	Visual

4.2.2.4 Body, prior to assembling head assembly (see dwg. 8848625)

Categories	Defects	Method of Inspection
Critical: None defined		
Major		
	101. Weight of charge	Balance
	102. Liner fails to cover all holes	Visual
	103. Closing plug improperly sealed	Visual
	104. Closing plug above flush	Visual
Minor:		
	201. Foreign matter in charge	Visual

4.2.2.5. Assembly (see dwg. 8848625)

Categories	Defects	Method of Inspection
Critical: None defined		

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Major:

101.	Case thread damaged	Gage
102.	Any flash hole not completely sealed (see Note I)	Visual
103.	Head assembly loose or assembled to wrong end of primer body.....	Manual-Visual

Minor:

201.	Total length, maximum (max.)	Gage
202.	Diameter of flange, max.....	Gage
203.	Crimp of body to head missing or incomplete	Visual
204.	Marking missing, misleading unidentifiable	Visual
205.	Evidence of poor workmanship (see 3.6)	Visual

Note I. Sketches 1, 2 and 3 of Figure 1 indicate acceptable primers. Furthermore, a primer with holes displaying the conditions 4, 5 and 6 may be classed acceptable provided the visible liner is completely covered with paint or lacquer and not more than 20 percent of the holes have this condition.

4.2.2.6 Carton, prior to sealing (see dwg. 8848889)

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
101.	Number of primers incorrect.....	Visual
Minor:		
201.	Any packing material missing.....	Visual
202.	Separator damaged to the extent that primers touch each other.....	Visual

4.2.2.7 Carton, sealed (see dwg. 8848889)

Categories	Defects	Method of Inspection
Critical:	None defined	

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Major: None defined

Minor:
201. Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal carton.....Visual
202. Carton torn, cut or punctured.....Visual
203. Marking misleading or unidentifiable .....Visual
204. Contents loose .....Manual

4.2.2.8 Sealed barrier bag (see dwg. 8848889)

Categories	Defects	Method of Inspection

Critical: None defined

Major:
101. Seal improper or incomplete .....Visual
102. Bag punctured, torn or cut .....Visual

Minor:
201. Marking misleading or unidentifiable .....Visual

4.2.2.9 Packing box, prior to sealing (see dwg. 8848888)

Categories	Defects	Method of Inspection

Critical: None defined

Major:
101. Carton missing.....Visual

Minor: None defined

4.2.2.10 Packing box, sealed (see dwg. 8848888)

Categories	Defects	Method of Inspection

Critical: None defined

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Major:

101.	Box broken, board loose or protruding nail	Visual
102.	Hardware or strapping, broken or loose	Visual-Manual

Minor:

201.	Car seal missing, unsealed or improperly positioned	Visual
202.	Hardware or strapping improperly engaged or mislocated	Visual
203.	Marking misleading or unidentifiable	Visual
204.	Contents loose (when tilted)	Manual

4.2.3 Testing

4.2.3.1 Head assembly

4.2.3.1.1 Non-functioning (see 3.3.1) Critical defect

4.2.3.1.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, a sample of 150 assemblies shall be randomly selected from each lot for the test. If any sample fails to comply with requirement the lot shall be rejected. The test shall be performed as specified in 4.3.1.1, using test equipment specified in 4.2.4.

4.2.3.1.1.2 After three consecutive lots have met the criteria of 4.2.3.1.1.1, a sample of 50 assemblies shall be randomly selected from each lot for this test. If one or more primers fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.1.1, using test equipment specified in 4.2.4.

4.2.3.1.2 Functioning (see 3.3.2) Major Defect

4.2.3.1.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, two hundred and fifty samples shall be randomly selected from each lot for this test. If three or more primers fail to comply with the applicable requirements, the lot shall be rejected. If one or two defects are found,

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a second sample of five hundred primers shall be tested. If the combined number of defectives in both the first and second shall sample is three or more, the lot shall be rejected. The test shall be performed as specified in 4.3.1.2, using test equipment specified in 4.2.4.

4.2.3.1.2.2 After three consecutive lots have met the criteria of 4.2.3.1.2.1 a sample of 125 head assemblies shall be randomly selected from each lot for test. If 2 or more assemblies fail to comply with the specified requirement, the lot shall be rejected. If only one defective is found, a second sample of 250 head assemblies shall be randomly selected. If the combined number of defectives in both the first and second samples is 2 or more, the lot shall be rejected. The test shall be performed as specified in 4.3.1.2, using test equipment specified in 4.2.4.

4.2.3.2 Primer Assembly Functioning (see 3.4) Major defect.- Twenty-five samples shall be randomly selected from each lot for this test. If one or more assemblies fail to comply with the requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.2.

4.2.3.3 Moisture content of black powder (see 3.5) Major defect.-A sample of one primer assembly, prior to sealing, shall be randomly selected from each eight hours production. The black powder shall be removed and tested as specified in 4.3.3. Any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample.

4.2.3.4 Check test for deterioration of percussion elements-Major defect.-If the total elapsed time between original acceptance of the percussion element lot and the assembly of that lot into the primers exceeds two years or if the percussion elements have been subjected to adverse conditions for any period of time since previous test, the percussion element lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-20449 immediately before the percussion element lot is assembled into the primer head assembly. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the completed primer (see 6.5).

4.2.3.5 Heat seal test of sealed bag (see dwg. 8848889) Major defect.-Sampling, acceptance criteria and test procedures for this test shall be as prescribed in Specification MIL-P-116 for the "Heat Seal Test".

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4.2.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 and 4.2.3 (see 6.2).

4.3 Test methods and procedure

4.3.1 Head assembly

4.3.1.1 Non-functioning.-The head assembly shall be assembled to the applicable test fixture specified in 4.2.4 so that the firing pin will strike the percussion element. A steel ball weighing 7.94 plus or minus .02 ounces shall be dropped on the firing pin from a height of six (6) inches, minimum (min.). The height of drop shall be measured as the hieght from the top of the firing pin to the bottom of the ball. Any head assembly which functions shall be classed defective. Head assemblies used in this test shall not be returned to the lot.

4.3.1.2 Functioning.-The test shall be performed as specified in 4.3.1.1, using the same steel ball, except that the height of drop shall be 20 inches, max. Any head assembly which fails to function shall be classed defective.

4.3.2 Primer assembly functioning.-This test shall be performed at a Government proving ground. The sample primers shall be assembled to complete rounds for which the primer was designed and fired from a gun for which the round is standard.

4.3.3 Moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level B.-The primers shall be packaged in accordance with dwg. 8848889. Barrier bag protection is required (see 6.3).

5.1.2 Level C.-The primers shall be packaged in accordance with dwg. 8848889. The use of the barrier bag is not required.

5.2 Packing

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5.2.1 Packing Level C.-Primers packaged as prescribed in 5.1.1 or 5.1.2 shall be packed in accordance with dwg. 8848888.

5.3 Marking

5.3.1 Marking.-Exterior containers shall be marked in accordance with dwg. 8848888.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Level of preservation and packing required (see 6.3.

c. Data card shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

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c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envolope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial Equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204 "Inspection Equipment Design" for guidance. The item detail specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Level B packaging and packing.-In the event the expected storage time of the primers is in excess of 180 days,

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the barrier bag shall be used as specified on the applicable drawing. Expected storage time shall be determined by the procuring agency.

6.4 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the percussion elements or to the contractor assembling the percussion elements into the primer assemblies and shall not constitute a basis of rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the percussion element lots were under the jurisdiction of either contractor.

6.5 Check test for possible deterioration of M81 Primer.-The following tests are designed to prevent loading major ammunition items with primers which have deteriorated. The tests should be performed on each lot of M81 Primers which have been stored for more than two years or which have been subjected to any adverse conditions, however brief, within that period of time. If the specification for the ammunition item into which the M81 Primer is assembled, does not require the performance of the following check test, the contracting officer shall contractually provide for compliance of the check test requirements.

6.5.1 Head assembly functioning.-The sampling plan for this test shall be in accordance with Table II. The body and head assembly shall be separated in a manner that will not damage the head assembly to the extent that the validity of the test results would be questionable. The test shall be performed as specified in 4.3.1.2 using test equipment specified in 4.2.4. If the total number of defective primers is equal to or exceeds the appropriate rejection number the lot shall be rejected.

TABLE II

Sample	Sample Size	Cumulative Sample Size	Combined Sample Acceptance Number	Rejection Number
First	75	75	*	2
Second	75	150	0	3
Third	75	225	0	3
Fourth	75	300	1	4
Fifth	75	375	2	5
Sixth	75	450	2	5
Seventh	75	525	4	5

*No acceptance on first sample.

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6.5.2 Moisture content of black powder.-A sample of 25 primers shall be randomly selected from each lot for this test. The black powder shall be removed from each primer by any suitable means and the moisture content determined *in accordance with the method specified in Specification MIL-P-223. Composite samples shall not be used. If the moisture content of the powder tested from any primer exceeds 0.7 percent, the lot of primers shall be rejected.

*Since black powder is hygroscopic, it is advisable to place the powder in a dry air-tight container immediately upon removal.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1390-A089

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ARTILLERY PRIMERS
STANDARDS FOR PAINT CONDITIONS IN FLASH HOLES

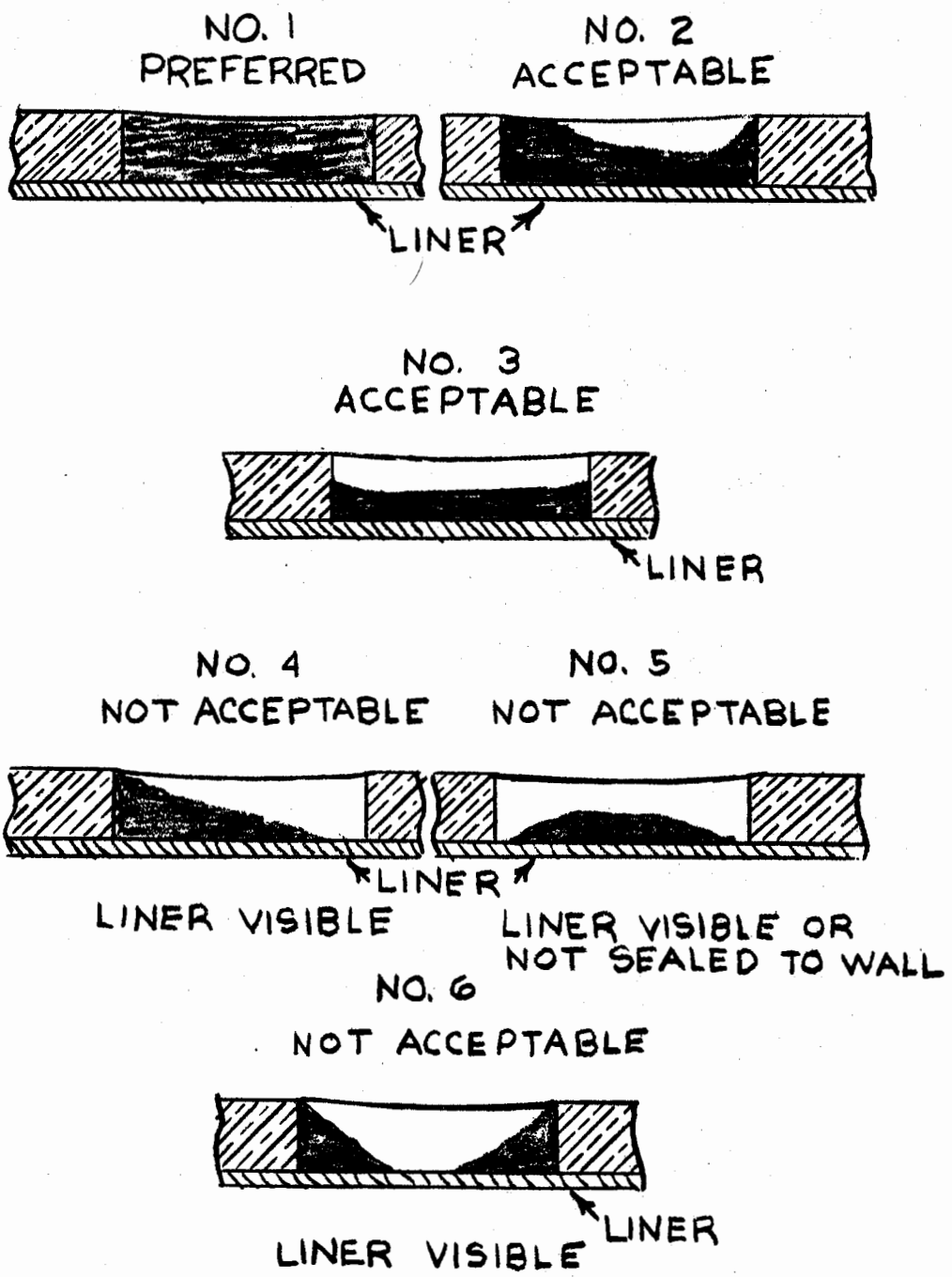


FIGURE 1

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